① $S \rightarrow AB$

$A \rightarrow aA \mid \varepsilon$

$B \rightarrow bB \mid c$

First(S) = { a, b, c }
First(A) = { a, ε }
First(B) = { b, c }

Follow (S) = { $ }
Follow (A) = { b, c }
Follow (B) = { $ }

② $E \rightarrow TR$

$R \rightarrow +TR \mid \varepsilon$

$T \rightarrow FY$

$Y \rightarrow *FY \mid \varepsilon$

$F \rightarrow (E) \mid id$

First (E) = { (, id }
First (R) = { +, ε }
First (T) = { (, id }
First (Y) = { *, ε)
First (F) = { (, id }

Follow(E) = {), $ }
Follow(R) = {), $ }
Follow(T) = { +,), $ }
Follow(Y) = { +,), $ }
Follow(F) = { *, +,), $)

(3) $S \rightarrow ABC$
$A \rightarrow a \mid \varepsilon$
$B \rightarrow b \mid \varepsilon$
$C \rightarrow c$

First(S) = { a, b, c }
First(A) = { a, ε }
First(B) = { b, ε }
First(C) = { c }

Follow(S) = { $ }
Follow(A) = { b, c }
Follow(B) = { c }
Follow(C) = { $ }

(4) $S \rightarrow aS \mid bA$
$A \rightarrow cA \mid d$

First(S) = { a, b }
First(A) = { c, d }

Follow(S) = { $ }
Follow(A) = { $ }

(5) $S \rightarrow (S)\,S \mid \varepsilon$
First(S) = { (, ε }
Follow(S) = {), $ }

(6) $S \rightarrow AB$
$A \rightarrow a \mid \varepsilon$
$B \rightarrow bC$
$C \rightarrow c \mid \varepsilon$

First(S) = { a, b }
First(A) = { a, ε }
First(B) = { b }
First(C) = { c, ε }

Follow(S) = { $ }
Follow(A) = { b }
Follow(B) = { $ }
Follow(C) = { $ }

(7)

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \varepsilon$

$F \rightarrow id \mid (E)$

First(E) = { id, (}

First(E') = { +, ε }

First(T) = { id, (}

First(T') = { *, ε }

First(F) = { id, (}

Follow(E) = {), $ }

Follow(E') = {), $ }

Follow(T) = { +,), $ }

Follow(T') = { +,), $ }

Follow(F) = { *, +,), $ }

(8)

$S \rightarrow AB$

$A \rightarrow Ba \mid \varepsilon$

$B \rightarrow b \mid \varepsilon$

First(S) = { b, a, ε }

First(A) = { b, a, ε }

First(B) = { b, ε }

CamScanner

Follow(S) = $\{\$\}$
Follow(A) = $\{b, a, \$\}$
Follow(B) = $\{a\}$

(9) $S \rightarrow ABC$
$A \rightarrow aA \mid \varepsilon$
$B \rightarrow bB \mid \varepsilon$
$C \rightarrow c \mid d$

First(S) = $\{a, b, c, d\}$
First(A) = $\{a, \varepsilon\}$
First(B) = $\{b, \varepsilon\}$
First(C) = $\{c, d\}$

Follow(S) = $\{\$\}$
Follow(A) = $\{b, c, d\}$
Follow(B) = $\{c, d\}$
Follow(C) = $\{\$\}$

(10) $S \rightarrow ACB$
$A \rightarrow aA \mid \varepsilon$
$B \rightarrow bB \mid \varepsilon$
$C \rightarrow c \mid d$

First(S) = {a, c, d, b}
First(A) = {a, ε}
First(B) = {b, ε}
First(C) = {c, d}

Follow(S) = {$}
Follow(A) = {c, d, b}
Follow(B) = {$}
Follow(C) = {b}

CamScanner

Ayesha Zubair

52916

CFG Lab 6

Task 1:

$S \rightarrow A\ B$
$A \rightarrow a\ A \mid \varepsilon$
$B \rightarrow b\ B \mid c$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 5
Enter productions:
S=AB
A=aA
A=$
B=bB
B=c

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c }
FOLLOW(S) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b c }
FOLLOW(B) = { $ }
Continue? (y/n): n

--------------------------------
Process exited after 16.25 seconds with return value 0
Press any key to continue . . .
```

Task 2:

$E \rightarrow T\ R$
$R \rightarrow +\ T\ R \mid \varepsilon$

$T \rightarrow F\ Y$

$Y \rightarrow *\ F\ Y \mid \varepsilon$

$F \rightarrow (\ E\ ) \mid id$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 8
Enter productions (Example: E=TR):
E=TR
R=+TR
R=$
T=FY
Y=*FY
Y=$
F=(E)
F=id

Find FIRST and FOLLOW of: F
FIRST(F) = { ( i }
FOLLOW(F) = { * + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: Y
FIRST(Y) = { * $ }
FOLLOW(Y) = { + $ ) }
Continue? (y/n): N

--------------------------------
Process exited after 120.9 seconds with return value 0
Press any key to continue . . .
```

Task 3:

$S \rightarrow A\ B\ C$

$A \rightarrow a \mid \varepsilon$

$B \rightarrow b \mid \varepsilon$

$C \rightarrow$ **c**

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 6
Enter productions (Example: E=TR):
S=ABC
A=a
A=$
B=b
B=$
C=c

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c }
FOLLOW(S) = { $ }
Continue? (y/n): n

--------------------------------
Process exited after 33.23 seconds with return value 0
Press any key to continue . . .
```

Task 4:

$$S \rightarrow a\ S \mid b\ A$$
$$A \rightarrow c\ A \mid d$$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 4
Enter productions (Example: E=TR):
S=aS
S=bA
A=cA
A=d

Find FIRST and FOLLOW of: A
FIRST(A) = { c d }
FOLLOW(A) = { $ }
Continue? (y/n): n

--------------------------------
Process exited after 44.51 seconds with return value 0
Press any key to continue . . .
```

Task 5:

$$S \rightarrow (S) S \mid \varepsilon$$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 2
Enter productions (Example: E=TR):
S=(S)S
S=$

Find FIRST and FOLLOW of: S
FIRST(S) = { ( $ }
FOLLOW(S) = { $ ) }
Continue? (y/n): n

--------------------------------
Process exited after 10.35 seconds with return value 0
Press any key to continue . . .
```

Task 6:

$$S \rightarrow A\ B$$
$$A \rightarrow a \mid \varepsilon$$
$$B \rightarrow b\ C$$
$$C \rightarrow c \mid \varepsilon$$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 6
Enter productions (Example: E=TR):
S=AB
A=a
A=$
B=bC
C=c
C=$

Find FIRST and FOLLOW of: C
FIRST(C) = { c $ }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b }
Continue? (y/n): n

--------------------------------
Process exited after 39.21 seconds with return value 0
Press any key to continue . . .
```

Task 7:

$E \rightarrow T\ E'$

$E' \rightarrow +\ T\ E' \mid \varepsilon$

$T \rightarrow F\ T'$

$T' \rightarrow *\ F\ T' \mid \varepsilon$

$F \rightarrow id \mid (\ E\ )$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 8
Enter productions (Example: E=TE'):
E=TE'
E'=+TE'
E'=$
T=FT'
T'=*FT'
T'=$
F=id
F=(E)

Find FIRST and FOLLOW of: F
FIRST(F) = { id ( }
FOLLOW(F) = { * + $ ) }
Continue? (y/n): n

--------------------------------
Process exited after 13.87 seconds with return value 0
Press any key to continue . . .
```

Task 8:

S → A B

A → B a | ε

B → b | ε

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 5
Enter productions (Example: E=TR):
S=AB
A=Ba
A=$
B=b
B=$

Find FIRST and FOLLOW of: S
FIRST(S) = { b a $ }
FOLLOW(S) = { $ }
Continue? (y/n): n

--------------------------------
Process exited after 31.59 seconds with return value 0
Press any key to continue . . .
```

Task 9:

$S \rightarrow A\ B\ C$

$A \rightarrow a\ A \mid \varepsilon$

$B \rightarrow b\ B \mid \varepsilon$

$C \rightarrow c\ C \mid d$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 7
Enter productions (Example: E=TR):
S=ABC
A=aA
A=$
B=bB
B=$
C=cC
C=d

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c d }
Continue? (y/n): n

--------------------------------
Process exited after 58.09 seconds with return value 0
Press any key to continue . . .
```

Task 10:

$S \rightarrow A\ C\ B$

$A \rightarrow a\ A \mid \varepsilon$

$B \rightarrow b\ B \mid \varepsilon$

$C \rightarrow c \mid d$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 7
Enter productions (Example: E=TR):
S=ACB
A=aA
A=$
B=bB
B=$
C=c
C=d

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { b $ }
Continue? (y/n): n

--------------------------------
Process exited after 11.66 seconds with return value 0
Press any key to continue . . .
```